Ex:02

Learning and Assignment of IP Address manually to Computer

Aim:-

To learn Ip addressing and manually assign an Ip address to a computer. To verify the assigned Ip address using Linux networking commands.

Theory:-

An Ip address is a unique numerical address assigned to every device connected to a network. It enables communication between devices over a network. An IPv4 address consists of four octs seperated by dots (eg: 192.168.1.10)

In Linux, the IP commands is used to display and manage, Ip address, network interface routing tables and neighbouring devices. It is one of the most commonly used commands for network configuration and troubleshooting.

Commands and sample outputs.

1, Display all Ip address

Command:-

ip address show

Sample output:

Sample output:-

2: ens160:

inet 172.16.211.128/24

inet fe80::20c:29ff:fe75:9923

observation:-

Display the IpN4 and IPV6 addresses of all networks.

2. Display a specific network interface:-

Commands:-

ip addr show dev ens160

Sample output:-

2: ens160:-

inet 172.16.211.128/24

3, Display Routing Table:-

Command

ip route show

Sample output:-

default via 172.16.211.2 dev ens160

172.16.211.0/24 dev ens160

observation:-

Display the routing table and the default gateway.

4. Display Neighbours (ARP) Table:

Command:

ip neigh show

Sample output

172.16.211.1 REACHABLE

172.16.211.254 STABLE

observation:

Display the neighboring devices and their MAC address.

Result:

The Ip address was manually assigned and verified successfully using Linux networking Commands.